

Master's Comprehensive Examination

Part I

Theory (90 minutes)

December 2, 1995

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. Men and women arrive at a bank as independent Poisson processes, men with rate 10 per hour and women with rate 6 per hour.

(i) Find the probability that more than 70 men and women arrive in a four-hour period.

(ii) Find the expected number of men and women arrive in a working day of eight-hour period.

2. $X_1, \dots, X_4$ are i.i.d. $N(\theta, 100)$. We want to test

$$H_0 : \theta \leq 400 \quad \text{vs} \quad H_a : \theta > 400.$$

(i) Consider the test procedure with rejects H_0 if both X_1 and X_2 are greater than 410. What is the Probability(Type I error) of this test?

(ii) Give a better test (based on all the $X_1, \dots, X_4$) than that in (i) and having the same size.

3. Let Z denote a standard $N(0, 1)$ random variable, and let $\Phi(\cdot)$ and $\phi(\cdot)$ denote its c.d.f. and p.d.f., respectively. Find the mean and variance of $\Phi(Z)$ and $\phi(Z)$.

4. A random variable X has a gamma distribution with parameters α and β if its density function is

$$f(x; \alpha, \beta) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad x > 0.$$

(a) Find the moment generating function for X , and find $E(X^k)$.

(b) Let $X_1, X_2, \dots, X_n$ be independent with X_i having a gamma(α_i, β) distribution. Using moment generating functions, find the distribution of

$$\bar{X} = \frac{1}{n}(X_1 + X_2 + \dots + X_n).$$

5. Let X be the only observation from the density $f(x; \theta) = (\theta + 1)x^\theta I_{(0,1)}(x)$, where $\theta > -1$.

(i) In testing $H_0 : \theta \leq 1$ vs $H_a : \theta > 1$, find the power function and the size of the following test: reject H_0 iff $X > .5$.

(ii) Find an estimator of θ based on the method of moments.

(iii) Find a uniformly most powerful size α test of $H_0 : \theta \geq 2$ vs $H_a : \theta < 2$.

Master's Comprehensive Examination

Part II

Applied (90 minutes)

December 2, 1995

Instructions: Attempt any four out of the following five questions. Be sure to indicate which four questions you would like to have graded. Open book and open notes.

1. A researcher seeks to test a new reducing diet on 10 women selected at random with the results given in the table below.

<u>Woman</u>	<u>Weight before</u>	<u>Weight after</u>
1	140	132
2	145	140
3	156	155
4	158	155
5	162	158
6	189	192
7	195	185
8	210	186
9	264	238
10	322	160

The researcher carries out a two-sample t-test on these data.

(a) Explain why the researcher's choice of test is inappropriate here.

(b) Carry out an appropriate test to determine whether the diet helps to reduce weight. Can you conclude (at the 5% level of significance) that the diet is effective?

(c) Give a one-sided 80% confidence interval for the average weight loss of the form: the average weight loss is at least _____ pounds. State which average you are using.

2. The weight of fully grown hogs are recorded, along with the size of their birth litter, in order to see if the size of the litter affects their adult weights. Altogether 30 hogs were weighed. Of the 30 hogs, 5 came from litters of 4, 10 came from litters of 5, 10 came from litters of 6, and 5 came from litters of 7.

Two ANOVA tables are given below. The first ANOVA table is constructed by treating the problem as a difference between four groups. The second ANOVA table is constructed by regressing weight on litter size. (Assume that the distribution of weights for a given litter size is normal, that the variance of weights is the same for each litter size, and that all responses are independent.)

(1) Complete the ANOVA tables on the next page.

(2) What hypotheses do the F-statistics test in the first table? The second table?

(3) Test at the $\alpha = .05$ level whether the simple linear regression model is appropriate. That is, is there sufficient evidence to suggest that the linear regression model does not hold?

ANOVA I

<u>Source</u>	<u>df</u>	<u>SS</u>	<u>MS</u>	<u>F</u>	<u>R²</u>
Model		3000			
Error					
Total		8200			

ANOVA II

<u>Source</u>	<u>df</u>	<u>SS</u>	<u>MS</u>	<u>F</u>	<u>R²</u>
Regression		2000			
Error					
Total					

3. For each of the following situations, assume that an Analysis of Variance is to be performed. List the sources of variation, the degrees of freedom and the expected mean square. Be sure to indicate which sources of variation are fixed and which are random.

(a) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. Each of 6 state experiment stations has agreed to grow the 3 varieties. Each station will homogenize a randomly selected tomato of each variety, determine the percent acid content, and submit this one measurement (for each variety) to the experimenter.

(b) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. From a single experiment station he obtains a bushel of each variety. For each variety he then selects 6 tomatoes at random from the bushel and records the percent acid content of each of the selected tomatoes.

(c) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. In a greenhouse, 4 benches were used. Placed on each bench was a single plant from each variety. From each plant 2 fruit were selected and the percent acid content of each fruit was recorded.

(d) An experimenter wants to compare the percent acid content of 3 varieties of tomatoes. He finds 6 home gardeners who are growing variety A, 6 who are growing variety B, and 6 who are growing variety C. Each gardener will supply the experimenter with 2 tomatoes and the experimenter will then record a measure of percent acid content for each tomato.

4. In a certain city the ethnic distribution of its residents, based on the 1990 census, is:

White	65%
Black	15%
Hispanic	10%
Other	10%

In 1995 a sample of 1000 citizens of this city results in the following ethnic distribution:

White	600
Black	100
Hispanic	100
Other	200

(a) Determine if there has been a shift in the ethnic distribution between 1990 and 1995.

(b) Construct a 95% confidence interval for the proportion of Hispanics in 1995; in 1990.

5. The probability, p , that a given die comes up as 6 is to be tested.

$$H_0: p = 1/6$$

$$H_a: p > 1/6$$

The die is to be rolled 600 times and X represents the number of times the die comes up 6. H_0 will be rejected if X exceeds a critical value C .

(a) Find C such that the test will have a 5% level of significance.

(b) For the value of C in part (a), find the power of the test against the alternative $p = 2/6$.

(c) If $X = 150$, what would be the observed level of significance (p -value)?

Master's Comprehensive Examination

Part I

Theory (90 minutes)

March 4, 1995

Instructions: Answer all four questions. Open book and open notes.

1. Let X be a random variable whose density is

$$f_X(x) = \begin{cases} cx^2 & \text{if } 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

- a) Find c .
- b) Find $P\{\frac{1}{4} \leq X < \frac{3}{4}\}$.
- c) Find the cdf $F_X(x)$ of X .

2. Don Mattingly's batting average is .350. Assuming he continues to bat as he has in the past, what is the approximate probability that he will get at least 200 hits next season assuming he has 500 at bats. (Assuming no walks in the 500 at bats.) Answer must be a number and not a sum of terms.

3. Let X_1, X_2 be a random sample of size 2 from a gamma distribution with parameters $\alpha = 3$ and θ . That is,

$$f_X(x; \theta) = \frac{e^{-x/\theta} x^2}{\Gamma(3)\theta^3}, \quad x > 0.$$

- a) Find a sufficient statistic T .
 - b) Is $\log_e(X_1 + X_2)$ a sufficient statistic?
4. a) For the same model as in question 1, find the maximum likelihood estimator of θ .
- b) Find the maximum likelihood estimator of e^θ .

Master's Comprehensive Examination

Part II

Applied (90 minutes)

March 4, 1995

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

1. In a large legislative body, 50% of the members are age 35 to 55, 25% are age 56 to 64, and 25% are age 65 or older. In a sample of 20 people chosen to form a key committee, 5 people are age 35-55, 8 people are age 56-64, and 7 people are age 65 or older. It is claimed that the 20 people were chosen completely at random from the legislative body. Carry out a test of this claim.

2. Let P denote the probability that a spun penny will come up tails. We seek to test the Null Hypothesis: $P = 0.5$, versus the Alternative Hypothesis $P > 0.5$. We are going to spin the penny 100 times and observe X , the total number of tails in the 100 spins. The test will reject the Null Hypothesis if X exceeds some critical number C .

a) Find C such that the test will have a 5% level of significance.

b) For the value of C in part (a), find the Power of the test against the alternative $P = 0.60$.

c) If $X = 60$, what would be the observed level of significance (p -value)?

3. A researcher is investigating the relation between the amount of money spent on probation services (per offender placed on probation) by a county, and the total number of crimes committed in those counties. For a sample of 20 counties, the mean amount spent on probation services is \$1,000, with a standard deviation of \$500. For these same counties, the mean number of crimes committed is 100 (per community), with a standard deviation of 30 crimes. The correlation between number of crimes committed and the amount spent on probation services is -0.60 . (a negative correlation).

a) The least-squares estimate of the average number of crimes committed in counties that spend \$1800 on probation is _____

b) The estimate of the standard deviation of the number of crimes committed in communities that spend \$1800 on probation per offender is _____

c) The 95% confidence interval for the average number of crimes committed in communities that spend \$1800 on probation is _____

4) Suppose that we have a chemical process in which a feed stock is pretreated in one of several ways to form an intermediate, and then the intermediate is processed in a second stage at one of two specific temperatures. Suppose there are 4 specific methods of pretreatment and 5 batches are prepared by each method. From each of the 20 batches we take 12 samples and treat 6 at one temperature and the other 6 at the other temperature. The goal of the study is to determine how long the feed stock will remain good under the different pretreatment-temperature processes. It is realized that each batch cannot be pretreated under ideal controlled conditions.

- Identify the response variable and the different factors.
- For each factor, state whether it is a treatment or a blocking factor, and whether it is fixed or random. Briefly justify your answer.
- For each pair of factors, state whether they are crossed or nested. Draw a data display.
- Write the full model for this example which accounts for all the possible effects. Clearly identify the indices.

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- Explain why the researcher's choice of test is inappropriate here.
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